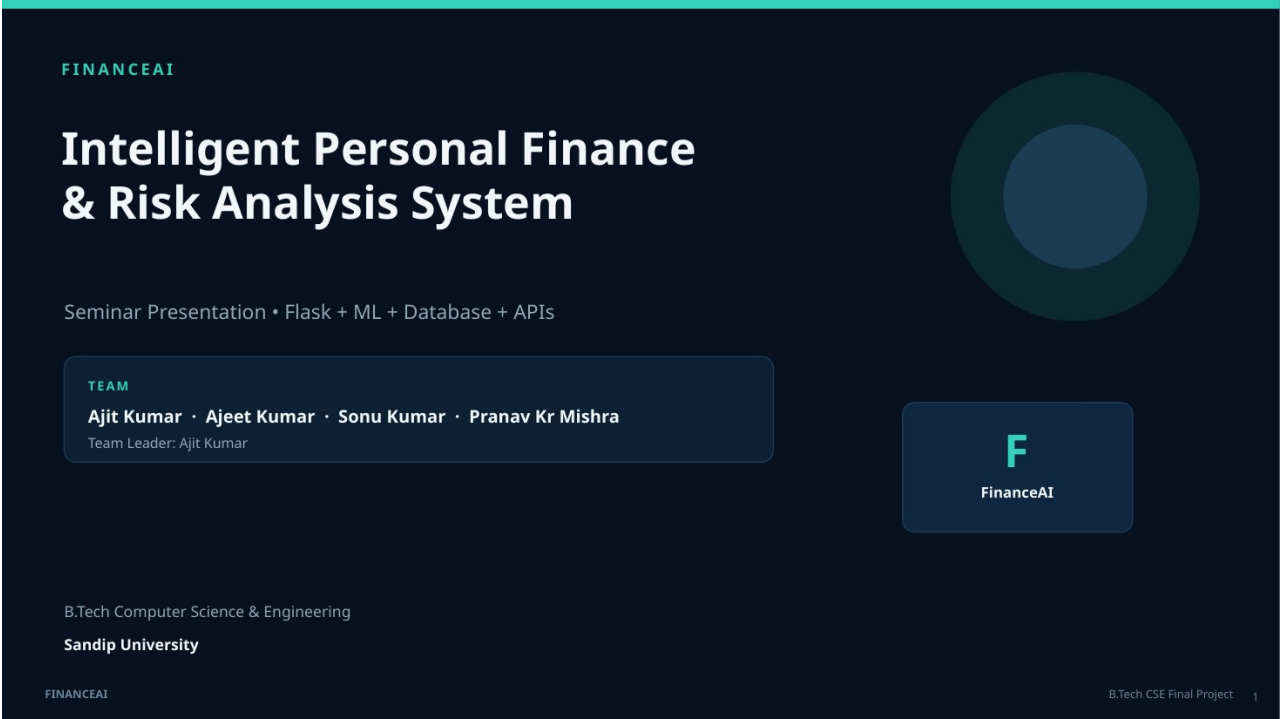


FINANCEAI SEMINAR SCRIPT

Intelligent Personal Finance & Risk Analysis System

Updated and aligned to the uploaded 19-slide seminar presentation



Team	Ajit Kumar • Ajeet Kumar • Sonu Kumar • Pranav Kr Mishra
Team Leader	Ajit Kumar
Program	B.Tech Computer Science & Engineering
University	Sandip University

Use this as a speaking guide - do not read every line mechanically. Keep eye contact, point to the slide, and use the transitions exactly where needed.

Presentation Flow & Speaker Handoffs

Recommended total seminar duration: approximately 16-20 minutes including the two short live-demo moments. Adjust according to faculty time. This timing is a delivery suggestion, not a project metric.

Slides	Speaker	Topic	Suggested time
1-5	Ajit Kumar	Opening, problem, solution, architecture	~5 min
6-8	Ajeet Kumar	Personal Finance + short demo	~3 min
9-11	Sonu Kumar	Credit Stress ML + explainability	~3.5 min
12-13	Pranav Kr Mishra	Loan Risk + limitations/demo	~2.5 min
14-19	Ajit Kumar (+ slide 17 with Pranav)	Application, backend, UX, responsible use, future, conclusion	~5 min

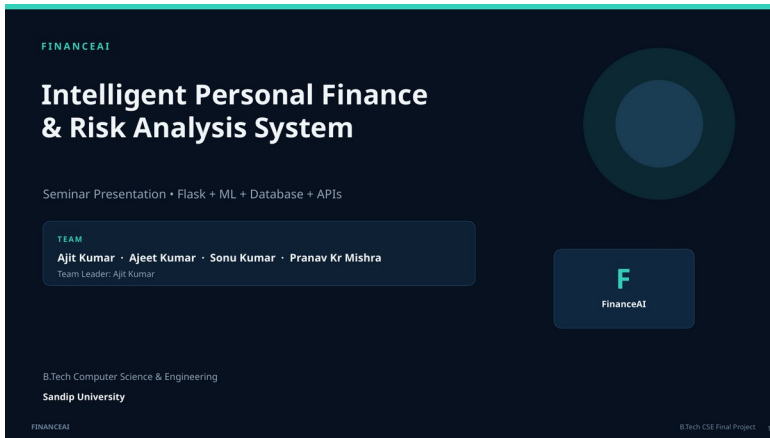
Before You Start

- Keep the project already running before the seminar starts and open the Home page in the browser.
- Keep sample input ready for Personal Finance and Loan Risk so the live demo does not become a typing session.
- When one member hands over to another, the next speaker should already be ready and should begin with "Thank you".
- Do not call Credit Stress or Loan Risk a real approval model. The deck explicitly labels them experimental/proxy-based.
- If the live demo fails, continue with the slide explanation and say you will show the saved result/history screen instead.

SLIDE 01 | FINANCEAI

Intelligent Personal Finance & Risk Analysis System

Seminar Presentation • Flask + ML + Database + APIs

**SPEAKER****Ajit Kumar****TIME****45-60 sec****FOCUS****FINANCEAI**

What to say

Good morning respected sir, ma'am and everyone present here.

My name is Ajit Kumar, and I am the team leader of our project.

Today, our team is going to present our final project, FinanceAI - Intelligent Personal Finance and Risk Analysis System.

Our team consists of four members: Ajit Kumar, Ajeet Kumar, Sonu Kumar, and Pranav Kr Mishra.

FinanceAI is basically a Flask-based integrated financial analytics application.

Is project me humne Personal Finance analysis, Machine Learning, authentication, database, APIs, analysis history aur explainable risk-analysis modules ko ek single application ke andar integrate kiya hai.

Hamare project ka main purpose sirf prediction karna nahi hai.

Hamara goal financial data ko understandable, explainable aur useful insights me convert karna hai.

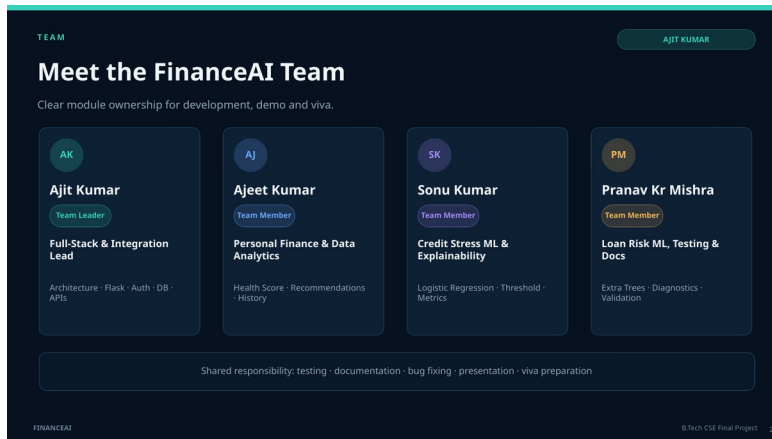
Presenter cues

- Start with a smile and eye contact. Do not look at the screen while saying your name and project title.
- Pause for one second after saying the full project name, then explain in one line what FinanceAI is.

SLIDE 02 | TEAM

Meet the FinanceAI Team

Clear module ownership for development, demo and viva.



SPEAKER

Ajit Kumar

TIME

45 sec

FOCUS

TEAM

What to say

Before moving further, I would like to introduce our team and responsibilities.

Main Ajit Kumar, Team Leader hoon, aur maine mainly overall architecture, Flask backend, authentication, database integration, APIs aur final module integration par work kiya hai.

Ajeet Kumar Personal Finance and Data Analytics part explain karenge.

Sonu Kumar Credit Stress Machine Learning and Explainability module handle kar rahe hain.

And Pranav Kr Mishra Loan Risk, testing and documentation module explain karenge.

Testing, debugging, documentation review aur final presentation ham sabki shared responsibility hai.

Presenter cues

- Point briefly to each team member card while naming responsibilities.
- Make the ownership clear, but also emphasize shared testing, documentation and presentation responsibility.

SLIDE 03 | WHY FINANCEAI

The Problem We Wanted to Solve

Financial data exists — but understanding it, acting on it and presenting risk responsibly is difficult.

WHY FINANCEAI

THE PROBLEM WE WANTED TO SOLVE

Financial data exists — but understanding it, acting on it and presenting risk responsibly is difficult.

- 01. Fragmented financial data**
Income, expenses, savings, debt and credit indicators are usually viewed separately.
- 02. Hard to interpret**
Raw numbers do not directly tell a user what is healthy, weak or urgent.
- 03. Black-box risk models**
Risk scores can be misunderstood when model limitations are hidden.

OUR GOAL Build one explainable platform that combines financial-health guidance, experimental ML risk analysis, authentication, history and APIs — without pretending experimental models are real lending decisions.

FINANCEAI

©Tech CSE Final Project

SPEAKER

Ajit Kumar

TIME

75-90 sec

FOCUS

WHY FINANCEAI

What to say

Now I would like to explain the problem that motivated us to build FinanceAI.

Aaj almost har person ke paas financial information hoti hai, jaise monthly income, expenses, savings, loan payment, investment aur credit-related data.

But problem ye hai ki raw financial numbers ko samajhna easy nahi hota.

For example, agar kisi ki salary fifty thousand rupees hai, to sirf salary dekhkar hum ye decide nahi kar sakte ki uski financial condition healthy hai ya nahi.

Hume expenses, debt, emergency fund, savings aur investment bhi dekhna padega.

Second problem ye hai ki financial risk models kabhi-kabhi black-box hote hain.

Model ek score de deta hai, but user ko clearly nahi pata hota ki result ka meaning kya hai, kis basis par prediction hua, aur model ki limitations kya hain.

So our problem statement was: Can we build one system which combines explainable financial-health analysis, machine-learning risk analysis and a complete application backend?

Isi problem ko solve karne ke liye humne FinanceAI develop kiya.

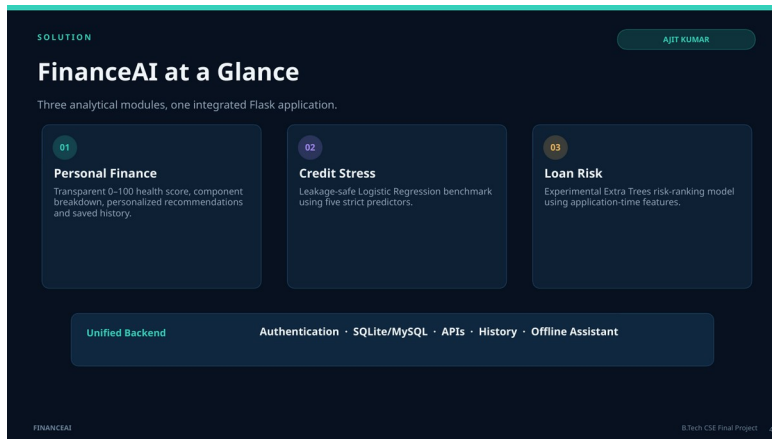
Presenter cues

- Use the salary example slowly; it makes the problem easy to understand.
- End with the problem statement as a question, then move to the solution.

SLIDE 04 | SOLUTION

FinanceAI at a Glance

Three analytical modules, one integrated Flask application.



SPEAKER

Ajit Kumar

TIME

45-60 sec

FOCUS

SOLUTION

What to say

FinanceAI ko humne three main analytical modules me divide kiya hai.

First module is Personal Finance.

Ye user ko zero to hundred ka Financial Health Score, component breakdown aur personalized recommendations deta hai.

Second module is Credit Stress.

Isme humne Logistic Regression based machine-learning benchmark use kiya hai.

Third module is Loan Risk.

Isme Extra Trees based experimental risk-ranking model use kiya gaya hai.

Aur in tino modules ko humne ek Flask application ke through integrate kiya hai.

Application me authentication, database, history, APIs aur offline FinanceAI Assistant bhi available hai.

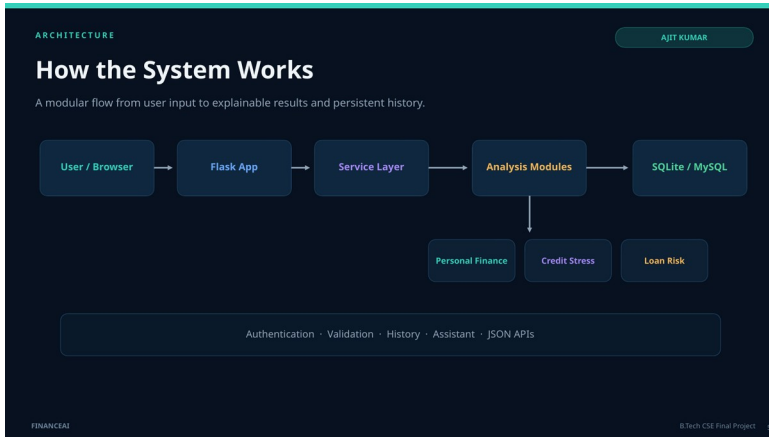
Presenter cues

- Count the three modules with your fingers: Personal Finance, Credit Stress, Loan Risk.
- Mention the unified Flask backend only after the three modules are clear.

SLIDE 05 | ARCHITECTURE

How the System Works

A modular flow from user input to explainable results and persistent history.



SPEAKER

Ajit Kumar

TIME

75 sec

FOCUS

ARCHITECTURE

What to say

Now let us understand how FinanceAI works technically.

Sabse pehle user browser ya API ke through application ko access karta hai.

User request Flask application par aati hai.

Flask layer authentication aur input validation handle karti hai.

Uske baad request service layer me jati hai.

Service layer decide karti hai ki Personal Finance engine, Credit Stress model ya Loan Risk model me se kaunsa module call karna hai.

Personal Finance module transparent rule-based scoring engine use karta hai.

Credit Stress Logistic Regression pipeline use karta hai.

Loan Risk Extra Trees pipeline use karta hai.

Final result SQLite ya MySQL database me save kiya ja sakta hai.

User later History page ke through past results dekh sakta hai.

Assistant latest saved Personal Finance result ko explain bhi kar sakta hai.

So overall architecture hai: User -> Flask -> Service Layer -> Analysis Engine or ML Model -> Database -> Result and History.

Transition: Ab Personal Finance module ko detail me explain karne ke liye main Ajeet ko invite karta hoon.

Presenter cues

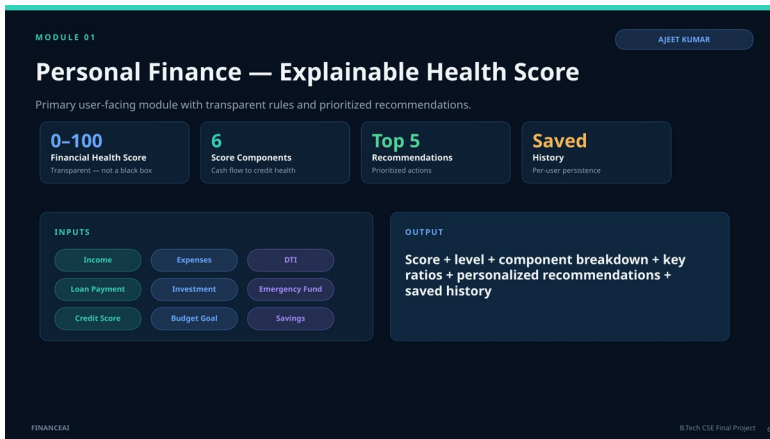
- Trace the architecture visually from left to right: User → Flask → Service Layer → Module → Database.
- At the end, clearly hand over to Ajeet.

HANDOVER: “Ab Personal Finance module ko detail me explain karne ke liye main Ajeet ko invite karta hoon.”

SLIDE 06 | MODULE 01

Personal Finance — Explainable Health Score

Primary user-facing module with transparent rules and prioritized recommendations.



SPEAKER

Ajeet Kumar

TIME

60 sec

FOCUS

MODULE 01

What to say

Thank you Ajit.

Now I am going to explain our first and primary user-facing module, which is Personal Finance.

Is module ka purpose hai user ki current financial condition ko understandable form me represent karna.

User kuch important inputs provide karta hai, such as monthly income, monthly expenses, debt-to-income ratio, loan payment, investment amount, emergency fund, credit score, budget goal, and actual savings.

In values ke basis par FinanceAI ek Financial Health Score from zero to hundred calculate karta hai.

Important point ye hai ki ye score black-box Machine Learning prediction nahi hai.

Ye transparent rule-based system hai.

Isliye hum clearly explain kar sakte hain ki total score kaunsa component kitna contribute kar raha hai.

Personal Finance module ka output sirf score nahi hai.

User ko health category, ratios, component breakdown aur personalized recommendations bhi milti hain.

Presenter cues

- Keep repeating that Personal Finance is transparent/rule-based, not a black-box ML prediction.
- On the score breakdown slide, make the six weights add up to 100 aloud.

SLIDE 07 | PERSONAL FINANCE

How the 100-Point Health Score is Built

Six transparent components make the score easy to explain in viva.



SPEAKER

Ajeet Kumar

TIME

60-75 sec

FOCUS

PERSONAL FINANCE

What to say

Now let us understand how the hundred-point score is calculated.

Health Score six components se milkar banta hai.

First is Cash Flow - maximum 25 points.

Second is Debt Health - maximum 20 points.

Third is Emergency Fund - maximum 20 points.

Fourth is Savings Progress - maximum 15 points.

Fifth is Investment - maximum 10 points.

And sixth is Credit - maximum 10 points.

So total becomes 100 points.

After calculating the score, system user ko health bands me classify karta hai.

85 to 100 Excellent, 70 to 84 Good, 55 to 69 Moderate, 40 to 54 Weak, and below 40 Critical.

Example integration test me score 77 out of 100 aaya, jo Good category hai.

Along with score, recommendation engine weak areas ko identify karta hai.

Agar expenses high hain to spending reduce karne ki recommendation, emergency fund low hai to emergency savings build karne ka suggestion, aur debt-to-income ratio high hai to additional debt avoid karne ka recommendation mil sakta hai.

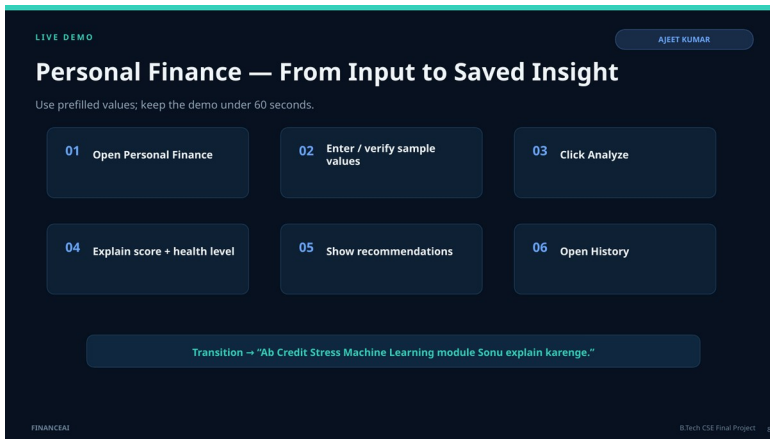
Presenter cues

- Keep repeating that Personal Finance is transparent/rule-based, not a black-box ML prediction.
- On the score breakdown slide, make the six weights add up to 100 aloud.

SLIDE 08 | LIVE DEMO

Personal Finance — From Input to Saved Insight

Use prefilled values; keep the demo under 60 seconds.



SPEAKER

Ajeet Kumar

TIME

45-60 sec demo

FOCUS

LIVE DEMO

What to say

Now I would like to show a small live demo.

Sabse pehle hum Personal Finance page open karte hain.

Yahan user monthly income, expenses, debt-to-income ratio, loan payment, investment, emergency fund, credit score, budget goal aur savings enter karta hai.

Now I click on Analyze.

System Financial Health Score generate karta hai.

Yahan health level show ho raha hai.

Neeche component-wise score show ho raha hai.

Aur recommendations section me prioritized actions show ho rahe hain.

After saving the result, ye analysis History page me available ho jata hai.

So Personal Finance module is explainable, transparent and user-focused.

Transition: Ab Credit Stress Machine Learning module Sonu explain karenge.

Presenter cues

- Use prefilled values. Avoid typing during the demo.
- Show only input → Analyze → score → recommendations → History; stop within one minute.
- End with the exact handoff to Sonu.

HANDOVER: “Ab Credit Stress Machine Learning module Sonu explain karenge.”

Credit Stress — Logistic Regression Benchmark

Leakage-safe classification for a financial-stress proxy target.

SPEAKER

Sonu Kumar

TIME

75 sec

FOCUS

MODULE 02

What to say

Thank you.

Now I am going to explain our Credit Stress Machine Learning module.

Is module me humne different classification models compare kiye.

Model comparison ke baad Logistic Regression champion model select hua.

But before discussing metrics, ek important point clear karna necessary hai.

Credit Stress module ka target is financial_stress_flag.

Ye actual credit default nahi hai.

It is a rule-based financial-stress proxy target.

Target leakage avoid karne ke liye humne intentionally sirf five strict predictors use kiye: Age, Dependents, Occupation, City Tier, and Desired Savings Percentage.

Income aur expense jaise features target ko directly construct karte the, isliye unhe strict ML predictor set se exclude kiya gaya.

Ye decision model ko scientifically safer banata hai.

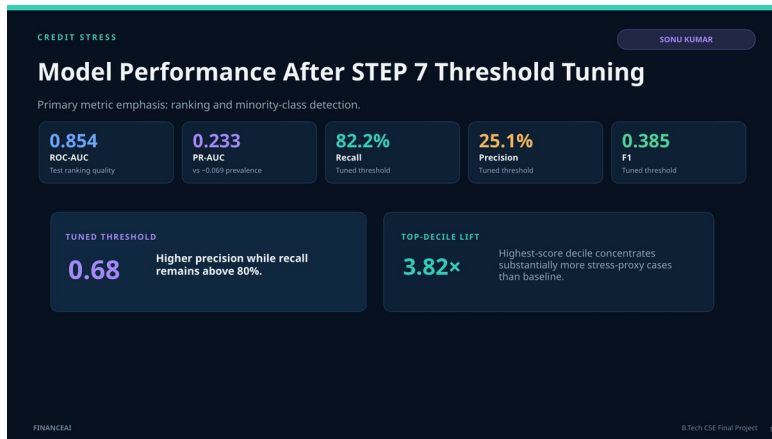
Presenter cues

- Say “proxy target” clearly and immediately clarify that it is not actual default.
- Emphasize that income/expense variables were excluded from the strict predictor set to reduce leakage.

SLIDE 10 | CREDIT STRESS

Model Performance After STEP 7 Threshold Tuning

Primary metric emphasis: ranking and minority-class detection.



SPEAKER

Sonu Kumar

TIME

75 sec

FOCUS

CREDIT STRESS

What to say

Now coming to the performance.

Champion model was Logistic Regression.

Test ROC-AUC approximately 0.854 tha.

PR-AUC approximately 0.233 tha.

Default threshold initially 0.50 tha.

STEP 7 me humne threshold tuning ki, aur recommended threshold became 0.68.

At this tuned threshold, Recall approximately 82.2 percent tha, Precision approximately 25.1 percent, and F1 score approximately 0.385.

Top-decile lift approximately 3.82 times tha.

Dataset imbalanced tha, jahan positive stress cases around 6.9 percent the.

Isliye humne accuracy ke place par PR-AUC, Recall, Precision aur F1 ko importance di.

Presenter cues

- Do not lead with accuracy. Start with ROC-AUC and PR-AUC because the positive class is imbalanced.
- When saying 0.68, call it the tuned decision threshold, not model accuracy.

SLIDE 11 | EXPLAINABILITY

Why Logistic Regression Fits This Module

Simple model, interpretable coefficients and a business-friendly threshold trade-off.



SPEAKER

Sonu Kumar

TIME

60-75 sec

FOCUS

EXPLAINABILITY

What to say

Logistic Regression choose karne ka ek additional advantage hai explainability.

Logistic Regression coefficients provide karta hai.

Agar coefficient positive hai, to wo feature model ka stress score increase karta hai.

Agar coefficient negative hai, to model score decrease hota hai.

Hum odds ratio bhi calculate kar sakte hain using: $\text{Odds Ratio} = \exp(\text{coefficient})$.

Isse prediction ka direction understand karna easier hota hai.

STEP 7 me threshold tuning ka purpose false positives aur missed positives ke beech better balance achieve karna tha.

Threshold 0.68 par precision improve hui aur recall abhi bhi 80 percent se above raha.

But ek limitation clearly mention karna important hai: This model does not predict actual credit default.

It predicts a financial-stress proxy.

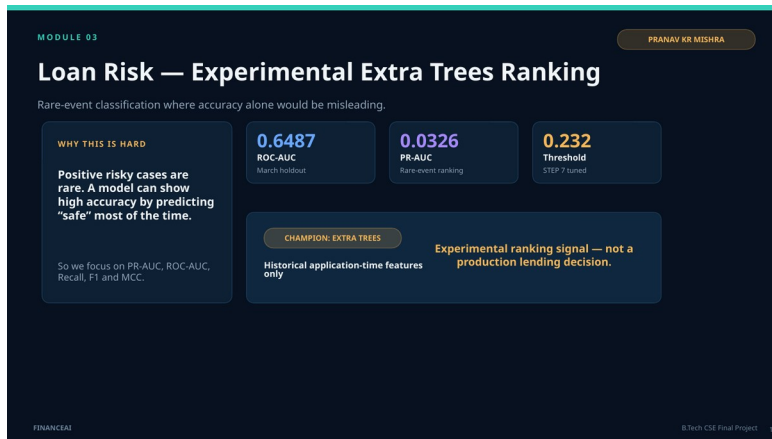
So FinanceAI is model ko credit-card approval ya rejection ke liye use nahi karta.

Presenter cues

- Use one simple sentence for coefficients: positive pushes score up, negative pushes score down.
- End by repeating: association is not causation; this is not an approval engine.

Loan Risk — Experimental Extra Trees Ranking

Rare-event classification where accuracy alone would be misleading.



SPEAKER

Pranav Kr Mishra

TIME

60-75 sec

FOCUS

MODULE 03

What to say

Thank you.

Now I am going to explain the Loan Risk module.

Loan Risk dataset me positive risky cases bahut kam the.

Ye severe class imbalance create karta hai.

Isliye agar hum sirf accuracy dekhte, to misleading result mil sakta tha.

For example, model almost sabko safe predict karke bhi high accuracy show kar sakta hai.

Isliye humne PR-AUC, ROC-AUC, Recall, F1 aur MCC jaise metrics use kiye.

Different models compare karne ke baad Extra Trees champion model select hua.

Test ROC-AUC approximately 0.6487 tha.

Test PR-AUC approximately 0.0326 tha.

STEP 7 me recommended threshold approximately 0.232 select hua.

Presenter cues

- Explain why high accuracy can be misleading when risky cases are rare.
- Say Extra Trees is the champion model from the compared candidates, then report the metrics exactly as on the slide.

SLIDE 13 | LOAN RISK

Safety, Input Validation & Demo

Weak rare-event performance is communicated honestly.

LOAN RISK

Safety, Input Validation & Demo

Weak rare-event performance is communicated honestly.

- 01 Rare positives**
Positive risky cases are scarce, so model quality is limited and highly sensitive to validation design.
- 02 Limited maturity**
Historical observation window is limited, especially for "Current" loans.
- 03 Complete input required**
API rejects incomplete application-time feature vectors instead of silently inventing values.

DEMO FLOW Load sample → Run score → Explain threshold → Show warning → Emphasize "experimental ranking only"

FINANCEAI | © Tech CSE Final Project | 13

SPEAKER

Pranav Kr Mishra

TIME

75 sec + demo

FOCUS

LOAN RISK

What to say

However, model ki limitation important hai.

Positive cases rare hain.

Historical observation window limited hai.

Aur tuned threshold ke baad bhi model performance real lending use ke liye strong enough nahi hai.

Isliye humne system me Loan Risk ko clearly experimental risk-ranking model ke roop me label kiya hai.

FinanceAI Loan Risk model ko automatic approval or rejection ke liye use nahi karta.

Ek aur important safety feature ye hai ki Loan Risk API complete application-time feature vector demand karti hai.

Agar required input missing hai, system silently median ya default value use karke misleading prediction nahi karta.

Instead, incomplete input ko reject karta hai.

Demo: Loan Risk page par sample input load karte hain.

Output me experimental risk score show hota hai.

Threshold ke comparison me class label generate hota hai.

Saath me warning bhi show hoti hai ki ye autonomous lending decision nahi hai.

Transition: Ab final backend features, future scope aur conclusion ke liye main presentation wapas Ajit ko hand over karta hoon.

Presenter cues

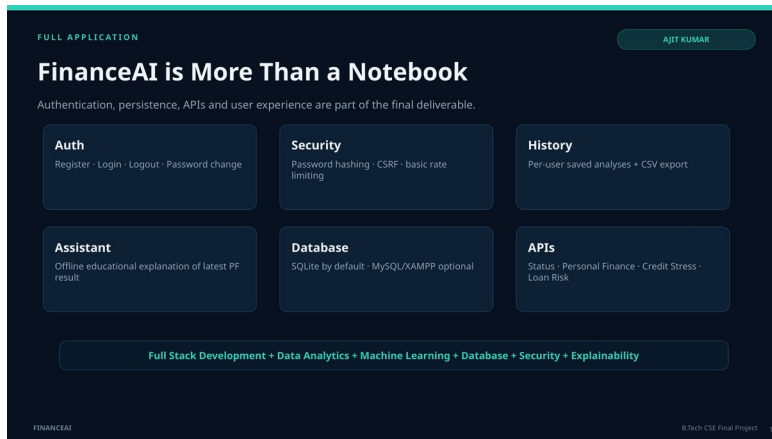
- Be confident when discussing limitations; honesty strengthens the project.
- For the demo, show sample → score → threshold → warning.
- Hand the presentation back to Ajit at the end.

HANDOVER: *“Ab final backend features, future scope aur conclusion ke liye main presentation wapas Ajit ko hand over karta hoon.”*

SLIDE 14 | FULL APPLICATION

FinanceAI is More Than a Notebook

Authentication, persistence, APIs and user experience are part of the final deliverable.



SPEAKER

Ajit Kumar

TIME

60 sec

FOCUS

FULL APPLICATION

What to say

Thank you Pranav.

Ab main explain karna chahunga ki FinanceAI sirf Machine Learning notebooks ka collection nahi hai.

Humne complete working Flask application develop kiya hai.

Application me user registration, login and logout, password change, secure password hashing, CSRF protection, basic login rate limiting, analysis history, CSV export aur offline FinanceAI Assistant available hain.

Har authenticated user ki analysis history separately maintain hoti hai.

Project JSON APIs bhi expose karta hai.

So our final project combines: Full Stack Development + Data Analytics + Machine Learning + Database + Security + Explainability.

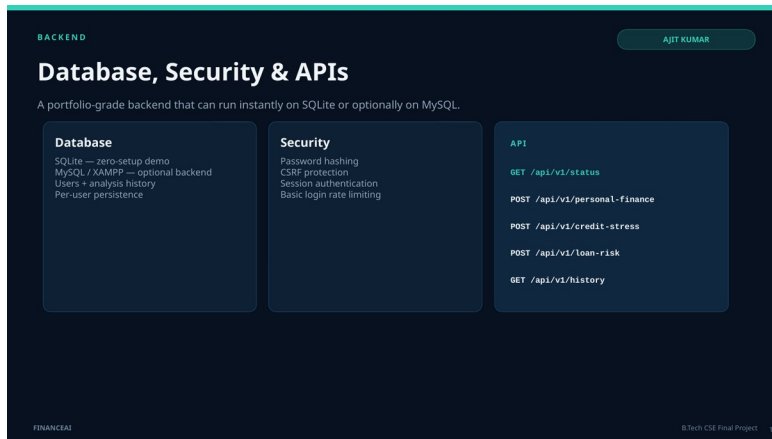
Presenter cues

- This slide proves the project is an application, not only notebooks.
- Group the features into: authentication, security, persistence/history, assistant, APIs.

SLIDE 15 | BACKEND

Database, Security & APIs

A portfolio-grade backend that can run instantly on SQLite or optionally on MySQL.



SPEAKER

Ajit Kumar

TIME

60 sec

FOCUS

BACKEND

What to say

For database, FinanceAI supports two options.

First is SQLite.

SQLite zero-setup hai, isliye instant local demo ke liye useful hai.

Second is MySQL, which can be used with XAMPP.

Database me user records aur analysis history persist hoti hai.

Security side par humne password hashing, CSRF protection, session authentication and basic login throttling implement kiya hai.

Project APIs me examples hain: GET /api/v1/status, POST /api/v1/personal-finance, POST /api/v1/credit-stress, POST /api/v1/loan-risk, GET /api/v1/history.

Ye APIs project ko future integration ke liye useful banati hain.

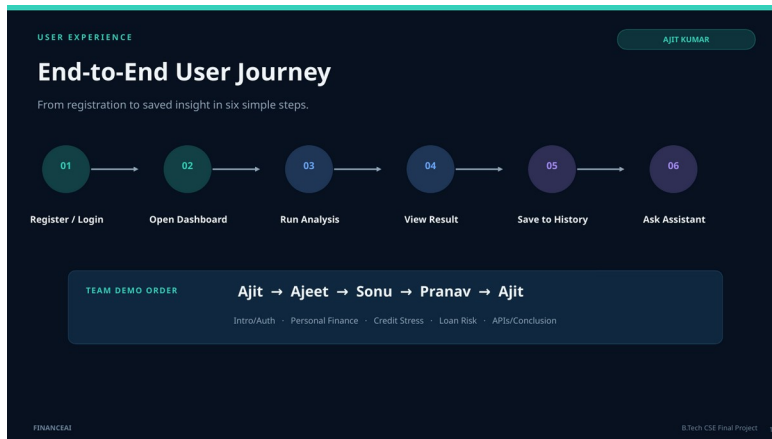
Presenter cues

- Explain SQLite first as the zero-setup demo option, then MySQL/XAMPP as optional.
- Read only 2-3 API examples aloud; point to the rest on the slide.

SLIDE 16 | USER EXPERIENCE

End-to-End User Journey

From registration to saved insight in six simple steps.



SPEAKER

Ajit Kumar

TIME

45-60 sec

FOCUS

USER EXPERIENCE

What to say

Complete user flow ko agar simple terms me samjhein, to six main steps hain.

First, user registers or logs in.

Second, user dashboard open karta hai.

Third, required analysis module select karta hai.

Fourth, input provide karke analysis run karta hai.

Fifth, system result show karta hai aur relevant result history me save hota hai.

And sixth, user Assistant ke through apne Personal Finance result ko simple language me samajh sakta hai.

So humne complete user journey ko simple, integrated aur reusable rakha hai.

Live flow: Home page → Login → Dashboard → module demos → History → Assistant → /api/v1/status.

Presenter cues

- Walk through the six-step journey quickly.
- If doing a full live flow, keep the browser tabs ready in this exact order.

SLIDE 17 | RESPONSIBLE USE

What FinanceAI Does — and Does Not Do

Model limitations are part of the product, not hidden footnotes.

The slide is titled "What FinanceAI Does — and Does Not Do" under the heading "RESPONSIBLE USE". It features a dark blue background with two main columns. The left column, titled "FINANCEAI DOES", lists four capabilities with green checkmarks: "Explain personal financial health", "Generate educational recommendations", "Demonstrate ML risk ranking responsibly", and "Show thresholds, metrics and limitations". The right column, titled "FINANCEAI DOES NOT", lists four limitations with red X marks: "Approve / reject credit cards", "Approve / reject loans", "Claim proxy labels are real defaults", and "Provide professional financial advice". At the bottom, a yellow box states "Responsible analytics = useful output + honest boundaries". The slide also includes a top right button "PRANAV + AJIT" and a footer with "FINANCEAI" and "©Tech CSE Final Project 17".

SPEAKER

Pranav Kr Mishra + Ajit Kumar

TIME

60 sec

FOCUS

RESPONSIBLE USE

What to say

FinanceAI clearly distinguishes what the system can do and what it cannot do.

FinanceAI can: explain personal financial health, generate educational recommendations, demonstrate experimental risk ranking, and show model metrics and limitations.

But FinanceAI does not: approve or reject a credit card, approve or reject a loan, claim that proxy labels are real defaults, or provide professional financial advice.

Humne intentionally model limitations ko hidden nahi rakha.

Hamare liye responsible analytics ka meaning hai: Useful output plus honest boundaries.

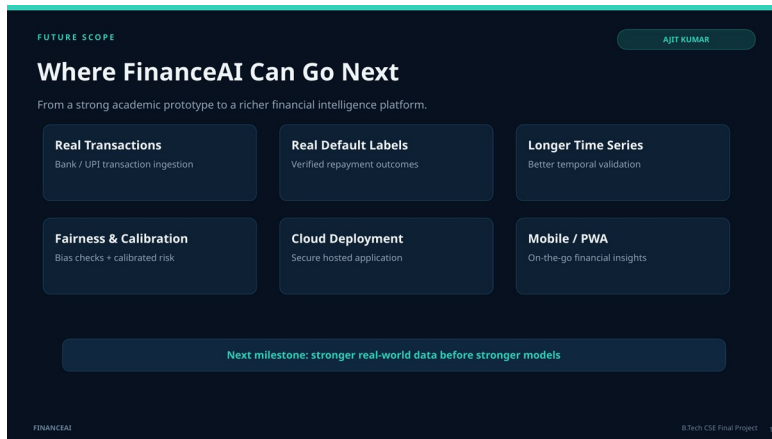
Presenter cues

- Split the slide into “does” and “does not”.
- Slow down for the “does not” list; these boundaries are important in viva.

SLIDE 18 | FUTURE SCOPE

Where FinanceAI Can Go Next

From a strong academic prototype to a richer financial intelligence platform.



SPEAKER

Ajit Kumar

TIME

60 sec

FOCUS

FUTURE SCOPE

What to say

Now coming to future scope.

Current project academic and portfolio stage par complete hai.

Future me FinanceAI ko real-world data ke saath improve kiya ja sakta hai.

Real bank or UPI transaction integration add ki ja sakti hai.

Credit Stress aur Loan Risk ke liye real verified repayment and default outcomes collect kiye ja sakte hain.

Longer historical time-series data use karke better temporal validation perform ki ja sakti hai.

Fairness and bias testing add ki ja sakti hai.

Risk probabilities ko properly calibrate kiya ja sakta hai.

Application ko secure cloud environment par deploy kiya ja sakta hai.

Mobile application ya Progressive Web App develop ki ja sakti hai.

Future FinanceAI Assistant ko bhi advanced user-approved financial context ke saath improve kiya ja sakta hai.

But our first priority in future would be: Better real-world data before building more complex models.

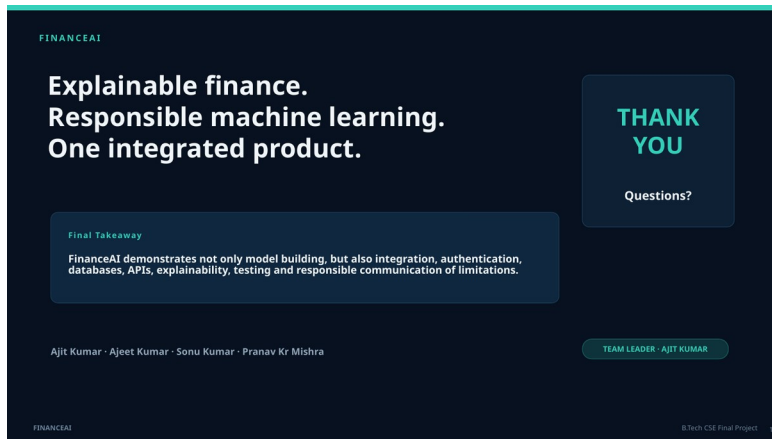
Presenter cues

- Future scope should sound realistic: better real-world data before more complex models.
- Mention real transactions, verified outcomes, temporal validation, fairness/calibration, cloud and PWA.

SLIDE 19 | FINANCEAI

Explainable finance. Responsible machine learning. One integrated product.

Final Takeaway

**SPEAKER****Ajit Kumar****TIME****75 sec****FOCUS****FINANCEAI**

What to say

To conclude our presentation

FinanceAI is not only a Machine Learning project.

It is an integrated financial analytics system.

Through this project, we worked on data understanding, data cleaning, EDA, feature engineering, Machine Learning, model evaluation, threshold tuning, explainability, Flask backend, authentication, database persistence, APIs, security, testing, and responsible communication of model limitations.

Our Personal Finance module provides an explainable Financial Health Score and recommendations.

Our Credit Stress module demonstrates Logistic Regression based financial-stress classification.

Our Loan Risk module demonstrates experimental Extra Trees risk ranking.

And all modules are integrated into one working Flask application.

Most importantly, we do not hide the limitations of our models.

FinanceAI demonstrates that a good AI or ML system is not only about getting a prediction.

It is also about making the result understandable, testable, secure and responsibly communicated.

Through this project, we learned both Machine Learning and real application integration.

On behalf of our entire team - Ajeet Kumar, Sonu Kumar, Pranav Kr Mishra and myself Ajit Kumar - thank you for giving us the opportunity to present our project.

Thank you.

We are ready for your questions.

Presenter cues

- Do not rush the conclusion. Summarize the integrated nature of the project.
- End with “Thank you. We are ready for your questions.” and stop speaking.

Likely Viva Questions - Short, Safe Answers

Q. FinanceAI me AI/ML exactly kahan use hua hai?

A. Credit Stress module Logistic Regression benchmark use karta hai, aur Loan Risk module Extra Trees based experimental risk-ranking model use karta hai. Personal Finance intentionally transparent rule-based scoring use karta hai.

Q. Personal Finance ko ML model kyon nahi banaya?

A. Current module ka goal explainability hai. Score six transparent components se banta hai, isliye user aur evaluator dono clear contribution dekh sakte hain.

Q. Credit Stress target actual default hai?

A. No. financial_stress_flag ek rule-based financial-stress proxy target hai. Presentation me explicitly clear hai ki ye observed default, delinquency, charge-off ya missed payment nahi hai.

Q. Target leakage kya problem thi?

A. Income aur expense variables proxy target ko directly construct karte the, isliye strict ML predictor set me unhe exclude kiya gaya taaki trivial label reconstruction aur leakage reduce ho.

Q. Credit Stress me accuracy kyon highlight nahi ki?

A. Positive stress cases around 6.9% the, isliye imbalance ke context me PR-AUC, Recall, Precision, F1 aur ranking metrics zyada informative hain.

Q. Threshold 0.68 ka kya meaning hai?

A. Ye tuned classification threshold hai. Presentation ke according is threshold par precision improve hui aur recall approximately 82.2% raha.

Q. Loan Risk model production lending ke liye ready hai?

A. No. Presentation me isse experimental ranking signal label kiya gaya hai. Rare positives, limited observation window aur modest PR-AUC ki wajah se ye autonomous approval/rejection ke liye use nahi hota.

Q. Extra Trees kyon?

A. Compared models me Extra Trees champion select hua. Deck isse historical application-time features par experimental risk ranking ke liye use karta hai.

Q. Security features kya hain?

A. Password hashing, CSRF protection, session authentication aur basic login rate limiting. User-specific analysis history bhi separately maintain hoti hai.

Q. SQLite aur MySQL dono kyon?

A. SQLite zero-setup local demo ke liye useful hai; MySQL/XAMPP optional backend ke roop me supported hai.

Q. Project ka strongest point?

A. Three analysis modules ko authentication, database persistence, history, APIs, assistant, security, explainability aur responsible limitations ke saath ek integrated Flask product me combine kiya gaya hai.

Q. FinanceAI kya nahi karta?

A. Credit card ya loan approve/reject nahi karta, proxy labels ko real defaults claim nahi karta, aur professional financial advice provide nahi karta.

60-Second Final Revision Before Seminar

- ✓ Opening: FinanceAI = integrated Flask financial analytics application with Personal Finance + Credit Stress + Loan Risk.
- ✓ Personal Finance: transparent 0-100 score; 25 Cash Flow + 20 Debt + 20 Emergency Fund + 15 Savings + 10 Investment + 10 Credit.
- ✓ Credit Stress: Logistic Regression; proxy target; strict 5 predictors; ROC-AUC 0.854; PR-AUC 0.233; threshold 0.68; Recall 82.2%; Precision 25.1%; F1 0.385; lift 3.82x.
- ✓ Loan Risk: Extra Trees; rare-event ranking; ROC-AUC 0.6487; PR-AUC 0.0326; threshold 0.232; experimental only.
- ✓ Backend: Flask + auth + hashing + CSRF + sessions + basic rate limiting + SQLite/MySQL + per-user history + APIs + offline assistant.
- ✓ Responsible use: no loan/credit approval, no professional financial advice, limitations are shown openly.
- ✓ Final line: "FinanceAI demonstrates that a good AI/ML system is not only about prediction; it is also about explainability, integration, security, testing and responsible communication."

ALL THE BEST FOR YOUR SEMINAR